Nonlinear inverse bremsstrahlung and the efficiency of gas lasers

Cooper K. Watson

October 20, 2026

Inverse bremsstrahlung absorption is nonlinear: a strong enough field reshapes the electron distribution that absorbs it, as Langdon showed, as Milder et al. observed directly and as Turnbull et al. have now measured in absorption [1, 5, 3, 4]. This note asks whether that nonlinearity can be used to make gas lasers more efficient. It locates the lasers and the available sources against the thresholds of the three nonlinearities, builds a kinetic equation for Langdon's and the gas's nonlinearity (the field, electron–atom and electron–ion collisions, and the electron–electron collisions that oppose the field, in the classical small-angle form of the collision integral whose quantum form is derived in [6]), checks it against Langdon's absorption factor, and applies it to the KrF laser gas, with growing and with beam-sustained electrons, and to the target that excimer drivers illuminate. Silin's regime is reached nowhere here. Every number computed here is checked numerically, as Appendix B describes. The results are in silico.

SI units are used, with electron energies ϵ and temperatures T in eV. The isotropic electron distribution $F(\epsilon)$ is normalized as $\int_0^\infty \epsilon^{1/2} F \, \mathrm{d}\epsilon = 1$, $\gamma = (2e/m)^{1/2}$, $v = \gamma \epsilon^{1/2}$, and 1 Td $= 10^{-21}$ V m^2. A field $E_0 \cos \omega t$ has the rms value $E = E_0/\sqrt{2}$, the quiver amplitude $v_0 = eE_0/m\omega$ and, for linear polarization, the intensity $I = c\epsilon_0 E_0^2/2$. Two fills of the KrF laser gas are used: the Ne fill, Ne/Kr/F_2 = 0.948/0.05/0.002 at 2 atm, and the Ar fill, Ar/Kr/F_2 = 0.597/0.40/0.003 at 1 atm, both at 300 K, with the LXCat cross sections BSR for neon, Biagi-v7.1 for krypton, IST-Lisbon for argon and SIGLO for F_2 [27]. The ions (cations and F^-) are taken to number 1.15 n_e at every n_e, the mean over the pump window that a kinetic model of an electron-beam-pumped laser gives for Kr/F_2 at 665 Torr with 2.8 Torr of F_2, taken for both fills. A revised form of that model gives 1.35, and with it krypton's share at beam-on density moves by at most 0.02 points.

Contents

1 The question and the answer **2**

2 Three nonlinearities of inverse bremsstrahlung **3**

3 The kinetic equation **4**

4 Checks of the kinetic model **6**

5 Where gas lasers sit **7**

6 Frequency as a lever on the distribution **9**

7 The self-sustained discharge **12**

8 What field pumping buys **14**

9 At the target **14**

10 Verdict and next steps **15**

A The electron–electron operator in energy **16**

B The checks **17**

1 The question and the answer

(1) A gas laser converts the energy of its electrons into excitation. The efficiency of that step depends on the shape of the electron distribution, because the excitation thresholds sit in its tail. Inverse bremsstrahlung (IB) is the way a field of frequency ω heats electrons, and its nonlinearity is that the heating reshapes the distribution. The question is whether a gas laser can be run in the nonlinear regime, by its own field or by an auxiliary one, so that the reshaped distribution puts more of the electrons' energy where the laser wants it.
(2) The answer has four parts.

- The lasing field of an excimer laser is linear. At a few saturation intensities (10 MW cm^{-2} for KrF, 30 for ArF) its intensity is $10^{3.8}$ to $10^{4.2}$ times below the intensity at which IB alone would supply the electrons' energy budget at $T_e = 1.5$ eV (at least 1.4×10^3 times for T_e from 1.08 to 2.04 eV), and further still below Langdon's and Silin's thresholds (Sec. 5). Of the lasers examined, the CO_2 laser is the exception: by an order-of-magnitude estimate its own field supplies about a tenth of its discharge electrons' budget at 100 MW cm^{-2}.

- An auxiliary field (microwave to far infrared) reaches only the nonlinearity of a weakly ionized gas, in which the field sets the electron temperature. It does so long before Langdon's flattening or Silin's saturation, which need tens and hundreds of times more intensity (33 to 80 and 380 to 920 times at the temperatures of Table 2).

- In that regime the frequency is a real lever on the distribution while electron–electron collisions are rare. A field far above the collision frequency roughly doubles krypton's share of the absorbed power at low power ($\times$2.0 to $\times$2.1 in the Ne fill at the power of 2 Td and $\times$2.1 to $\times$2.4 in the Ar fill at 6 Td, with growing or with beam-sustained electrons at a fixed electron density), and multiplies it by 12 to 32 in the Ar fill at 3 Td, where the DC share is only 1 to 5 %. Those collisions remove most of the lever at the densities of a pumped laser. Where the lever is large they do not spoil the high-frequency distribution: they pull both distributions towards the Maxwellian, and the DC one catches up. The advantage halves at a generalized Langdon parameter $\alpha_g = 0.41$ to 0.55. At the beam-on density of a pumped laser, 2.72×10^{21} m^{-3}, with the growth model or with beam-sustained electrons that recombine as Kr_2^+, it is 2 to 7 % where the field's power exceeds the beam's own and 10 to 14 % at a fixed beam current where the beam's is larger (up to 27 % at fixed n_e). Without recombination it is 1 to 18 %, and 3 to 7 % at a fixed beam current. With one electron per ion pair the beam-sustained model holds that density only with 13 times the beam power that gives it in a full kinetic model of the laser, and at the five times lower density it holds with that power

the advantage is up to ×1.24 at a fixed beam current and ×1.355 at fixed n_e (Sec. 6). The krypton share of a self-sustained discharge differs from its DC value by at most 1.5 percentage points in the Ne fill, and falls by up to ten points at high frequency in the Ar fill (Secs. 6 and 7).

- At the target the nonlinearity favours the short wavelengths of the gas lasers. At the quarter-critical conditions that NRL computed for 351, 248 and 193 nm, the Langdon factor of a homogeneous plasma is 0.94, 0.96 and 0.98 with the velocity-dependent logarithm. That factor alone raises the local absorption rate of KrF and ArF light by 2.3 and 3.8 % over 351 nm, on top of the much larger linear gain of the short wavelengths. Speckle and nonlocal transport, left out here, may change it (Sec. 9).

(3) The nonlinearity of IB is therefore not, by itself, a lever on the efficiency of a gas laser. What the computation does show is nearly frequency-blind and favourable. Field pumping of the beam-ionized Ne fill at beam-on density puts 0.30 to 0.35 of the absorbed energy into KrF^* where its power exceeds the beam's own, counted at the photon energy, depending on the power and on the electron-source model. The e-beam itself reaches 0.20 in the same accounting once its sub-excitation electrons are credited, so field pumping forms 1.5 to 1.8 times as much there, and the frequency moves that by −4 to +6 % (Sec. 8). The accounting leaves out electron collisions with Kr^* and KrF^*, which compete with the formation at beam-on density. Whether field pumping pays is a question of source efficiency, stability and chemistry, which this note does not address. The most useful application of the nonlinear absorption is an instrument: a THz pulse that sets T_e on demand and a transmission that reads it back.

2 Three nonlinearities of inverse bremsstrahlung

(1) *Langdon's nonlinearity: the distribution flattens.* In a fully ionized plasma with $\omega \gg \nu_{ei}$ the field heats electrons through their collisions with ions with the velocity-space diffusion coefficient $(v_0^2/6)\nu_{ei}(v)$ [1], and $\nu_{ei} \propto v^{-3}$ heats the slow electrons most. Electron–electron collisions restore the Maxwellian. Their competition is measured by Langdon's parameter

$$\alpha_L = \frac{Z_{\rm eff} v_0^2}{v_t^2}, \qquad v_t^2 = \frac{eT}{m}, \qquad Z_{\rm eff} = \frac{\langle Z^2 \rangle}{\langle Z \rangle}, \tag{2.1}$$

As α_L grows the distribution becomes flat-topped, which lowers the absorption by the factor f_L. Langdon found the absorption 12 % lower at $\alpha_L = 0.05$, 67 % of its Maxwellian value at 0.5 and 45 % in the limit [1]. This note takes the onset at $\alpha_L = 0.1$, where the self-similar solution of Sec. 4 gives $f_L = 0.83$ (Langdon's fit 0.82). Turnbull et al. measured the absorption of a 527 nm probe in gas-jet plasmas and found that it needs this factor [3]. The path-integrated f_L averaged 0.76 over all their data, and even over the eight pure-hydrogen shots it averaged 0.81 and fell as low as 0.74.
(2) *Silin's nonlinearity: the quiver outruns the electron.* When $v_0 \gtrsim v_t$ the collision samples the oscillating relative velocity and the absorption falls as $(v_t/v_0)^3$ up to a logarithm [16].
(3) *The nonlinearity of a weakly ionized gas: the field sets the temperature.* In a laser gas, n_e/N is 10^{-7} to 10^{-4} and electrons lose energy mainly to atoms and molecules. Elastic collisions remove only the fraction $2m/M$ of the energy per collision, 1.3×10^{-5} (Kr) to 5.4×10^{-5} (Ne), and the inelastic channels raise the effective fraction $\delta_{\rm eff}$ to 4×10^{-4} to 8×10^{-4} in the excimer fills (Sec. 5) and to about 10^{-2} in a CO_2 laser discharge. A field heats an electron at the rate $P = \sigma_{\rm abs} I$, where, in the Lorentz two-term approximation,

$$\sigma_{\rm abs} = \frac{e\,\mathrm{Re}\,\mu(\omega)}{c\epsilon_0} \xrightarrow{\omega \gg \nu} \frac{e^2}{mc\epsilon_0}\,\frac{\nu_{\rm eff}}{\omega^2}, \tag{2.2}$$

μ being the complex mobility. The field takes over the temperature once P exceeds the collisional loss $P_{\rm loss}(T_e)$ that holds T_e, at

$$I_{\rm heat} = \frac{P_{\rm loss}(T_e)}{\sigma_{\rm abs}(\omega)}, \qquad I_L(\alpha_L) = \frac{c\epsilon_0}{2}\Big(\frac{m\omega}{e}\Big)^2 \frac{\alpha_L\, eT}{mZ_{\rm eff}}, \qquad I_S = \frac{c\epsilon_0}{2}\Big(\frac{m\omega}{e}\Big)^2 \frac{eT}{m}, \tag{2.3}$$

the last two being the intensities of a given α_L and of $v_0 = v_t$. For $\omega \gg \nu$, Eqs. (2.2) and (2.3) give

$$\frac{I_L}{I_{\rm heat}} = \frac{\alpha_L\, eT\, \nu_{\rm eff}}{2Z_{\rm eff} P_{\rm loss}} = \frac{\alpha_L}{3Z_{\rm eff}\, \delta_{\rm eff}}, \qquad P_{\rm loss} = \tfrac{3}{2}\, eT\, \delta_{\rm eff}\, \nu_{\rm eff}. \tag{2.4}$$

A gas's electrons thus answer to an intensity $\alpha_L/3Z_{\rm eff}\delta_{\rm eff}$ times lower than the one that gives a plasma's distribution a given α_L, 33 to 80 times lower for $\alpha_L = 0.1$ at the temperatures of Table 2. $I_{\rm heat}$ is the threshold that Ginzburg and Gurevich called the plasma field: below it a wave crosses a weakly ionized plasma without changing it appreciably [29, 30]. In the gas, the nonlinearity is that the distribution, and with it $\nu_{\rm eff}$ and the absorption, follows the field. Its shape depends on ω: a DC field heats each electron in proportion to $1/\nu_m(\epsilon)$, a field with $\omega \gg \nu_m$ in proportion to $\nu_m(\epsilon)$ (Sec. 3).

(4) *The generalized Langdon parameter.* In every case the departure from the Maxwellian is decided by the field's heating against the Maxwellization by electron–electron collisions, while in a gas the energy dependence of ν_m and the inelastic losses also set its form. Define

$$\alpha_g = \frac{P_{\rm abs}}{eT_e\, \nu_{ee}}, \qquad \nu_{ee} = 2.91\times10^{-12}\, n_e\, \ln\Lambda\, T_e^{-3/2}\ {\rm s}^{-1} \quad (n_e \text{ in m}^{-3},\ T_e \text{ in eV}), \tag{2.5}$$

with $P_{\rm abs}$ the absorbed power per electron and ν_{ee} the electron collision rate of the NRL formulary [28]. For the Maxwellian of a fully ionized plasma the IB heating is $P = (e^2E^2/m\omega^2)\,\nu_T L$ with $\nu_T = (4\sqrt{2\pi}/3)\, n_i Z^2 e^4/[(4\pi\epsilon_0)^2 m^{1/2}(eT)^{3/2}]$ [7]. In practical units $\nu_T = 2.906\times10^{-12}\, n_i Z^2 T^{-3/2}$, the coefficient of ν_{ee}, so

$$\alpha_g = \frac{\alpha_L}{2}\,\frac{L}{\ln\Lambda}, \tag{2.6}$$

and Langdon's α_L is $2\alpha_g$ when the two logarithms are equal. The same α_g applies to a gas, where $P_{\rm abs}$ comes mostly from collisions with atoms.

3 The kinetic equation

(1) The isotropic part of the distribution obeys, in the two-term approximation and per electron,

$$\epsilon^{1/2}\frac{\partial F}{\partial t} = -\frac{\partial G}{\partial \epsilon} + S[F], \qquad G = G_{\rm field} + G_{\rm el} + G_{ee}, \tag{3.1}$$

with S the inelastic collisions (and the source or growth term of item (6)). The field flux is $G_{\rm field} = -D_{\rm field}F'$ with

$$D_{\rm field} = \frac{\gamma^2}{3}\, E^2\epsilon^{3/2}\,\frac{\nu}{\nu^2+\omega^2}, \qquad \nu = \nu_a + \nu_c, \tag{3.2}$$

where $\nu_a = N\gamma\epsilon^{1/2}\sigma_m$ is the momentum-transfer frequency on atoms and ν_c that on ions. At $\omega = 0$ this is the DC term of Hagelaar and Pitchford [9], $D = (\gamma/3)(E/N)^2\epsilon N/\sigma_m$, and for $\omega \gg \nu$ it is $(\gamma^2/3)(E/\omega)^2\epsilon^{3/2}\nu$, which is Langdon's operator with the velocity-space diffusion coefficient $(v_0^2/6)\nu$ [1]. The two-term expansion needs a small anisotropy, so it cannot describe Silin's regime.

At every scan point of Secs. 6 and 7 the electrons' oscillation velocity $eE_0/m(\omega^2+\nu_m^2)^{1/2}$, with the Maxwellian $\langle\nu_m\rangle$ on atoms, is at most $0.13\,v_t$. The elastic flux on atoms is $G_{\rm el}=-\gamma\epsilon^2 N\sigma_\epsilon(F+T_gF')$ with $\sigma_\epsilon=\sum_j x_j(2m/M_j)\sigma_{m,j}$.

(2) *Ions.* A singly charged ion has the Coulomb momentum-transfer cross section

$$\sigma_c(\epsilon)=\pi\Big(\frac{e}{4\pi\epsilon_0\,\epsilon}\Big)^2 L(\epsilon),\qquad L=\tfrac12\ln\Big[1+\Big(\frac{b_{\max}}{b_{\min}}\Big)^2\Big], \tag{3.3}$$

with $b_{\max}=v/(\omega^2+\omega_p^2)^{1/2}$ and $b_{\min}$ the larger of $e^2/4\pi\epsilon_0 mv^2$ and $\hbar/2mv$. For $\omega\gg\omega_p$, $b_{\max}$ is the IB cutoff v/ω, set by the laser frequency rather than the plasma frequency as Turnbull et al. found the absorption to need [3]. Below ω_p it becomes the screening length v/ω_p. Then $\nu_c=\pi n_i Z^2\gamma(e/4\pi\epsilon_0)^2 L\,\epsilon^{-3/2}$, and for $\omega\gg\nu_c$ Eq. (3.2) becomes, with $E^2/\omega^2=(v_0^2/v_t^2)\,T/\gamma^2$ and $n_iZ=n_e$,

$$D_{\rm IB}=\frac{\pi\gamma^3}{3}\,n_iZ^2\Big(\frac{e}{4\pi\epsilon_0}\Big)^2 L\,\frac{E^2}{\omega^2}=\frac{\pi\gamma}{3}\,\alpha_L\,T\,n_e\Big(\frac{e}{4\pi\epsilon_0}\Big)^2 L. \tag{3.4}$$

For a constant logarithm $D_{\rm IB}$ does not depend on ϵ: in energy, Langdon's operator is pure diffusion with a constant coefficient. The absorbed power is $P=\int D_{\rm IB}(-F')\,{\rm d}\epsilon=D_{\rm IB}F(0)$, proportional to the number of slow electrons, so $f_L=F(0)/F_M(0)$ at equal T, where F_M is the Maxwellian.

(3) *Langdon's v^5 distribution, exactly.* With heating alone the distribution obeys $\epsilon^{1/2}\partial_t F=D_{\rm IB}\partial_\epsilon^2 F$. Try $F=A(t)\exp[-\epsilon^{5/2}/b(t)]$. Then $\epsilon^{1/2}\partial_t F=\epsilon^{1/2}F(\dot A/A+\epsilon^{5/2}\dot b/b^2)$ and $\partial_\epsilon^2F=F[(25/4)\epsilon^3/b^2-(15/4)\epsilon^{1/2}/b]$. The powers $\epsilon^{1/2}$ and ϵ^3 match separately when

$$\frac{{\rm d}b}{{\rm d}t}=\frac{25}{4}\,D_{\rm IB},\qquad \frac{{\rm d}\ln A}{{\rm d}t}=-\frac{15}{4}\frac{D_{\rm IB}}{b}. \tag{3.5}$$

The distribution keeps the shape $\exp(-v^5)$ while it heats, Langdon's limit for $\alpha_L\to\infty$, and its absorption factor, that of the super-Gaussian $\exp[-(v/v_m)^M]$ with $M=5$ at the same density and energy, is $F(0)/F_M(0)=M(\pi/2)^{1/2}\Gamma(5/M)^{3/2}/[3^{3/2}\Gamma(3/M)^{5/2}]=0.4456$.

(4) *Electron–electron collisions.* The isotropic Rosenbluth operator [10] is the classical small-angle (Landau) form of the electron–electron collision integral, whose quantum form is derived in [6]. In energy it reads (Appendix A)

$$G_{ee}=-2a\,n_e\int_0^\infty \min(\epsilon,\epsilon')^{3/2}\big[F(\epsilon')F'(\epsilon)-F(\epsilon)F'(\epsilon')\big]\,{\rm d}\epsilon',\qquad a=\frac{e^2\gamma\ln\Lambda}{24\pi\epsilon_0^2}, \tag{3.6}$$

which is the form $-a\,n_e[3A_1F+2(A_2+\epsilon^{3/2}A_3)F']$ of Hagelaar and Pitchford [9], $A_1=\int_0^\epsilon u^{1/2}F$, $A_2=\int_0^\epsilon u^{3/2}F$, $A_3=\int_\epsilon^\infty F$. The kernel is symmetric and the bracket antisymmetric in (ϵ,ϵ'), so

$$\int_0^\infty G_{ee}\,{\rm d}\epsilon=0, \tag{3.7}$$

which, since $\int\epsilon\,\partial_\epsilon G\,{\rm d}\epsilon=-\int G\,{\rm d}\epsilon$, is the conservation of energy. Any $F\propto e^{-\epsilon/T}$ makes the bracket vanish. Far above the bulk, $\min=\epsilon'$, the friction is $2an_e\int\epsilon'^{3/2}F'\,{\rm d}\epsilon'=-3an_e$ and the diffusion is constant, so a fast electron loses energy at

$$\frac{{\rm d}(e\epsilon)}{{\rm d}t}=-\frac{3a\,n_e\,e}{\epsilon^{1/2}}=-\frac{e^4\ln\Lambda\,n_e}{4\pi\epsilon_0^2\,m\,v}, \tag{3.8}$$

the classical slowing-down rate of a fast electron by the field electrons, $7.7\times10^{-12}\,n_e\ln\Lambda\,\epsilon^{-3/2}$ s^{-1} in the units of Eq. (2.5) [28]. Discretized on the cell faces with face averages and differences of F, Eq. (3.7) holds to round-off and a sampled Maxwellian is an exact zero for every grid step.

(5) *The heated plasma at fixed α_L.* For a fully ionized plasma heated by IB, write $x = \epsilon/T$ and $F = T^{-3/2}\Phi(x)$ with T two thirds of the mean energy. When the shape is stationary while T rises, the flux in the frame that moves with T vanishes everywhere. Dividing by $2an_eT^{-3/2}$,

$$-\beta\, g(x)\Phi' - \left[\Phi' P(x) - \Phi\, Q(x)\right] = \lambda\, x^{3/2}\Phi, \qquad \beta = \frac{\alpha_L L}{4\ln\Lambda}, \tag{3.9}$$

with $P = \int \min(x,x')^{3/2}\Phi(x')\,\mathrm{d}x'$, $Q = \int \min(x,x')^{3/2}\Phi'(x')\,\mathrm{d}x'$, $g = L(xT)/L$ for a velocity-dependent logarithm (otherwise 1), and λ the heating rate $\dot{T}/T$ in e-e units. The ratio β follows from Eq. (3.4) and $2an_e = (4\pi/3)\gamma\ln\Lambda\, n_e(e/4\pi\epsilon_0)^2$. Integrating Eq. (3.9) over x, the e-e term drops out by Eq. (3.7) and $\int x^{3/2}\Phi = 3/2$ gives $\lambda = (2/3)\beta\int g(-\Phi')\,\mathrm{d}x$. For $\beta \to \infty$, Eq. (3.9) gives $\Phi \propto \exp(-cx^{5/2})$, the distribution of Eq. (3.5). For $\beta \to 0$ it gives the Maxwellian. By Eq. (2.6), $\beta = \alpha_g/2$ for the Maxwellian.
(6) *Method.* Equation (3.1) is solved on an energy grid with the two-term discretization of Hagelaar and Pitchford [9], extended by Eqs. (3.3) and (3.6). Three choices matter, and the first was forced by checks of Appendix B.

- A scheme whose inelastic gains are not exactly number-conserving, and whose LU solve replaces the top row by the normalization, satisfies the rows of the tail only to the accuracy of the largest F. Against an elimination that satisfies every row to round-off, such an LU solution is off by 2 % where $F/F_{\max}$ is 10^{-6} to 10^{-3} and by order one below. The solver therefore distributes each source cell's inelastic loss over its landing cells (number-conserving to round-off) and solves by Gaussian elimination from the top in ratio form, $c_i = F_i/F_{i-1}$, which satisfies every row but the lowest. That row is then exactly the particle balance, and the growth frequency is found as the root of it. Krypton's share of the power changes by 3×10^{-5} between the two schemes.

- The e-e coefficients are lagged and iterated with Anderson mixing [11] of $\ln F$, so that the tail converges in relative terms.

- A beam-sustained mode replaces the growth term by electrons injected with the sub-excitation spectrum of the beam's secondary electrons that a kinetic model of the laser uses (linear on 0 to 9.915 eV, mean 4.506 eV), lost by attachment and by recombination with a cross section $\propto 1/\epsilon$ (Sec. 6).

4 Checks of the kinetic model

(1) *Langdon.* Table 1 and Fig. 1 compare the solutions of Eq. (3.9) with equal logarithms ($\beta = \alpha_L/4$) to Langdon's fit $f_L = 1 - 0.553/[1 + (0.27/\alpha_L)^{0.75}]$, which he gives as good to ±0.005 [1] and Turnbull et al. use [3]. They agree within 1.8 % from $\alpha_L = 0.01$ to 100, the largest difference being at $\alpha_L = 0.5$. The kinetic solution is therefore the problem Langdon solved. The fitted super-Gaussian exponent depends on the fit, because the distribution is not exactly super-Gaussian. Matte's $m(\alpha_L) = 2 + 3/(1 + 1.66\alpha_L^{-0.724})$ [2, 4] lies between the exponents of two fits for $\alpha_L \le 5$ and within 0.09 of the three-parameter fit above. The energy relation of item (5) of Sec. 3 holds to 3×10^{-6}, and $\beta \to \infty$ reproduces the v^5 distribution's factor 0.4456 to 4×10^{-6}.
(2) *The gas.* With $n_e = 0$, no ions and the discretization of the first scheme of Sec. 3, the solver reproduces an independent two-term solver to 7×10^{-11}. A constant cross section in a DC field gives the Druyvesteyn distribution [12] to 1.2×10^{-7} in $\ln F$, a field with $\omega \gg \nu$ and elastic losses the Maxwellian of Margenau [13], with or without e-e collisions. The e-e flux conserves energy to

Table 1: The heated plasma at fixed α_L, Eq. (3.9) with $\beta = \alpha_L/4$, against Langdon's absorption fit and Matte's exponent. m is fitted in two ways: to $\ln\ln(\Phi(0)/\Phi)$ against $\ln x$ on $0.1 < x < 3$, and with three free parameters on $x < 3$.

α_L	f_L	Langdon's fit	ratio	m, $\ln\ln$	m, three-parameter	Matte's m
0.01	0.9564	0.9569	0.9994	2.12	2.03	2.06
0.03	0.9125	0.9108	1.0019	2.25	2.08	2.14
0.1	0.8292	0.8220	1.0088	2.51	2.21	2.31
0.3	0.7242	0.7126	1.0163	2.91	2.46	2.60
1	0.6065	0.5977	1.0147	3.53	2.97	3.13
3	0.5256	0.5250	1.0011	4.13	3.61	3.71
10	0.4761	0.4815	0.9888	4.62	4.31	4.28
30	0.4568	0.4627	0.9873	4.85	4.71	4.63
100	0.4491	0.4535	0.9904	4.95	4.91	4.83

1.4×10^{-17}, its drift on a fast electron is Spitzer's Eq. (3.8) to 2×10^{-5}, and two mixed Maxwellians relax to one. The Coulomb heating of a 3 eV Maxwellian is $(e^2E^2/m\omega^2)\nu_T L$ to 9×10^{-5}, once the half cell next to $\epsilon = 0$ is accounted for. In a DC field e-e collisions turn the Druyvesteyn distribution, whose $(2/3)\langle\epsilon\rangle/T_{\rm slope}$ is 1.50 with $T_{\rm slope}$ fitted to $\ln F$ on $0.1\langle\epsilon\rangle < \epsilon < 4\langle\epsilon\rangle$, into a Maxwellian: the ratio is 1.0005 at $n_e/N = 10^{-2}$.

5 Where gas lasers sit

(1) Table 2 evaluates Eq. (2.3) for the fields of three lasers and for auxiliary sources heating the Ne fill, with Maxwellian electrons, the losses of the LXCat sets, and the ions of Eq. (3.3). Every beam-on row has $T_e = 1.5$ eV, between the temperature of a Maxwellian power balance at beam-on density, 1.08 eV, and that of the beam-sustained electrons of Sec. 6 at beam-on density, 1.88 (Ar fill) and 2.04 eV (Ne fill), and $n_e = 2.72\times10^{21}$ m^{-3}. The afterglow rows have 0.63 eV and 10^{20} m^{-3}. The ArF fill is Electra's, $Ar/F_2 = 0.997/0.003$ at 0.82 atm [14]. The CO_2 row uses a CO_2:N_2:He $= 1:1:8$ mixture at 1 atm, $T_e = 1$ eV and $n_e = 10^{19}$ m^{-3} with the IST-Lisbon cross sections [27], and singly charged cations only ($Z_{\rm eff} = 1$). Langdon's and Silin's intensities are left out where $\nu_m > \omega$, where the electrons do not quiver freely. Figure 2 shows the thresholds against frequency.

Table 2: Thresholds of the three nonlinearities, Eq. (2.3), in W cm^{-2}. $u = \hbar\omega/eT_e$; ν_m/ω with the Maxwellian $\langle\nu_m\rangle$; $P_{\rm loss}$ in eV s^{-1} per electron; Langdon at $\alpha_L = 0.1$ with $Z_{\rm eff} = 1.15$ (1 for the CO_2 discharge); Langdon and Silin left out where $\nu_m > \omega$.

field, gas	T_e	n_e (m^{-3})	u	ν_m/ω	$P_{\rm loss}$	$I_{\rm heat}$	I_L	I_S
KrF, 248 nm, Ne fill	1.5	2.7×10^{21}	3.3	2.0×10^{-4}	1.1×10^{9}	7.1×10^{10}	5.7×10^{12}	6.5×10^{13}
ArF, 193 nm, ArF fill	1.5	2.7×10^{21}	4.3	1.4×10^{-4}	1.6×10^{9}	1.7×10^{11}	9.3×10^{12}	1.1×10^{14}
CO_2, 10.6 μm, discharge	1	1.0×10^{19}	0.12	0.0097	3.5×10^{10}	9.6×10^{8}	2.4×10^{9}	2.4×10^{10}
2.45 GHz, afterglow	0.63	1.0×10^{20}	1.6×10^{-5}	32	3.6×10^{8}	2.8×10^{2}	–	–
170 GHz, afterglow	0.63	1.0×10^{20}	0.0011	0.46	3.6×10^{8}	1.4×10^{3}	4.7×10^{4}	5.4×10^{5}
0.67 THz, afterglow	0.63	1.0×10^{20}	0.0044	0.12	3.6×10^{8}	1.5×10^{4}	7.3×10^{5}	8.4×10^{6}
1 THz, beam-on	1.5	2.7×10^{21}	0.0028	0.24	1.1×10^{9}	5.2×10^{4}	3.9×10^{6}	4.5×10^{7}
28.3 THz, beam-on	1.5	2.7×10^{21}	0.078	0.0086	1.1×10^{9}	3.9×10^{7}	3.1×10^{9}	3.6×10^{10}

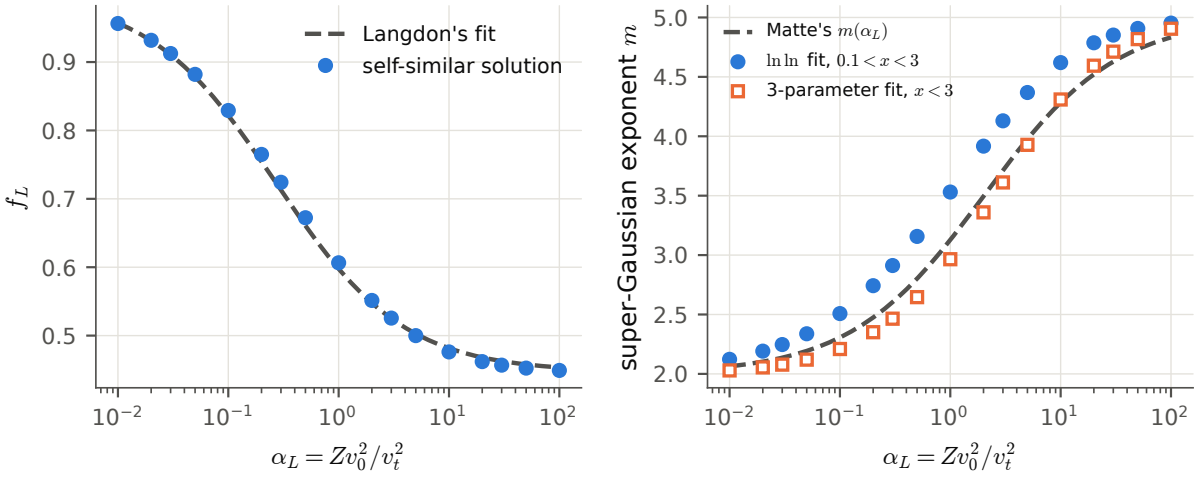


Figure 1: Left: the absorption factor of the heated plasma against Langdon's fit. Right: fitted super-Gaussian exponents against Matte's.

(2) *The lasers' own fields.* An amplifier extracts at a few saturation intensities. For KrF, $I_{\text{sat}} = h\nu(A+Q)/\sigma$ is 1.0 and 1.2 MW cm^{-2} in the two fills, with the cross section 2.6×10^{-16} cm^2 and radiative rate of Kannari et al. [31] and the quench rates of a KrF kinetic network at each fill's composition. Electra's ArF amplifier saturates at 10 MW cm^{-2} [14]. At 10 MW cm^{-2} the KrF field is 7.1×10^3 times below I_{heat} in the Ne fill and 1.5×10^4 times in the Ar fill, and at 30 MW cm^{-2} the ArF field is 5.8×10^3 times below it. These margins depend on T_e through the inelastic losses: for T_e from 1.08 to 2.04 eV they are $10^{3.2}$ to $10^{5.0}$. They also rest on the classical kernel at $u = \hbar\omega/eT_e = 3.3$ and 4.3, where it is not validated (the quantum absorption on these atoms, from exact phase shifts of model potentials, is 10 to 28 % below it at $u = 4.5$, for 300 K electrons at 28.3 THz, and its size for 5.0 and 6.4 eV photons is not known). The excimer lasers' own fields are linear by every measure. The CO_2 laser's field supplies a tenth (0.10) of the energy budget of its discharge electrons at 100 MW cm^{-2}, and all of it at $I_{\text{heat}} = 9.6 \times 10^8$ W cm^{-2}. That is an order-of-magnitude estimate. It assumes Maxwellian electrons at 1 eV, which at $n_e/N = 4 \times 10^{-7}$ nothing Maxwellizes, and discharge electrons present while the pulse passes, which depends on the timing of the amplifier. Of the lasers examined it is the one in which the lasing field can reach the gas nonlinearity. Lasing is known to perturb CO_2-laser discharges, the optogalvanic effect, but through the vibrational kinetics of the laser levels [26]. No study was found of the field's IB heating of the discharge electrons as an efficiency or stability term.

(3) *Auxiliary fields.* Heating the Ne fill's electrons takes 1.4×10^3 W cm^{-2} at 170 GHz and 1.5×10^4 W cm^{-2} at 0.67 THz in the afterglow, and 5.2×10^4 W cm^{-2} at 1 THz beam-on, within reach of gyrotron beams [32, 8]. Such a beam heats only its absorption length, 2.5 mm at 170 GHz and 2.6 cm at 0.67 THz in the afterglow and 1.1 mm at 1 THz beam-on. Where $\nu_m < \omega/2$, Langdon's $\alpha_L = 0.1$ needs 62 to 80 times I_{heat} in the Ne fill beam-on, 33 to 51 times in its afterglow, and 37 and 54 times at the laser frequencies of the Ar and ArF fills. Silin's $v_0 = v_t$ needs a further $Z_{\text{eff}}/\alpha_L = 11.5$ times. For $\omega \gg \nu_m$ the ratio is Eq. (2.4), $\alpha_L/3Z_{\text{eff}}\delta_{\text{eff}}$, with $\delta_{\text{eff}} = 4 \times 10^{-4}$ to 8×10^{-4}. In the excimer fills a field strong enough to matter is in the gas nonlinearity, and only there. The CO_2 discharge, whose vibrational losses make δ_{eff} large, has Langdon's intensity only 2.5 times its I_{heat}.

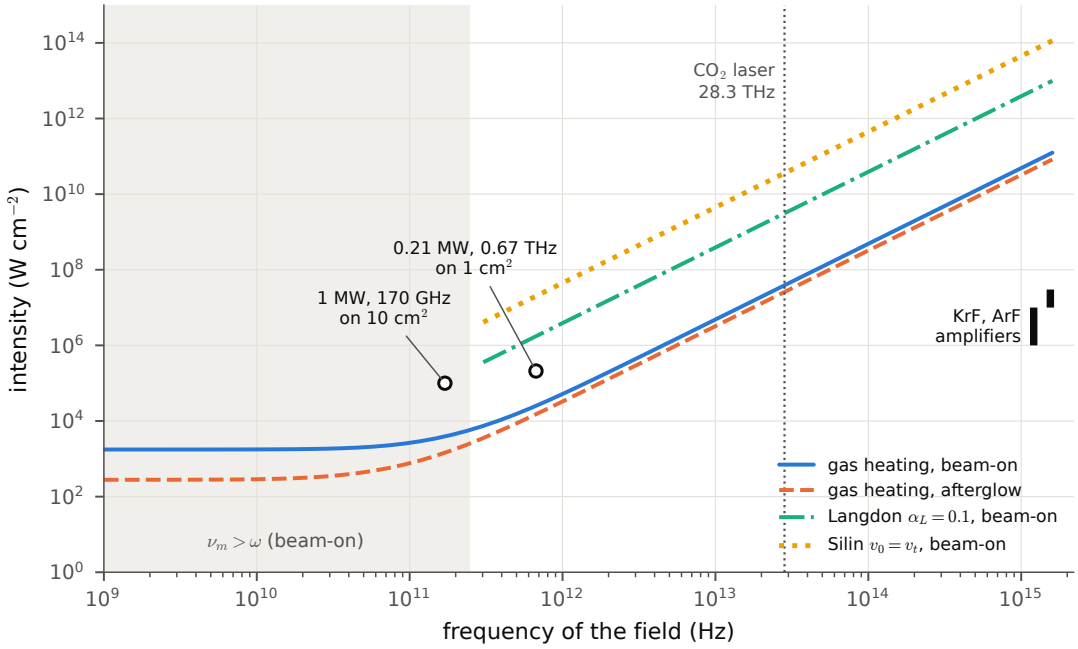


Figure 2: The three thresholds against frequency in the Ne fill: gas heating (I_{heat}) beam-on (T_e = 1.5 eV, $n_e = 2.72 \times 10^{21}$ m^{-3}) and in the afterglow (0.63 eV, 10^{20} m^{-3}), Langdon at $\alpha_L = 0.1$ and Silin, both beam-on and drawn only where $\nu_m < \omega$. Bars: KrF amplifiers at 1 to 10 and ArF amplifiers at 10 to 30 MW cm^{-2} (Sec. 5); dotted line: the CO_2-laser frequency 28.3 THz; circles: a 1 MW gyrotron at 170 GHz [32] and a 0.21 MW, 0.67 THz gyrotron [8], each on an assumed spot.

6 Frequency as a lever on the distribution

(1) Without e-e collisions a field above the collision frequency puts more of its power into krypton than a DC field of the same absorbed power: at the lowest power about twice as much in the Ne fill and up to 32 times as much in the Ar fill, where the DC share is 1.3 %. This section asks whether e-e collisions erase that. The mechanism is item (3) of Sec. 2. A DC field heats each electron in proportion to $1/\nu_m$, so in a gas whose ν_m rises with energy it heats the slow electrons most and gives the Druyvesteyn-like distribution, with a depleted tail. A field with $\omega \gg \nu_m$ heats in proportion to ν_m, which with elastic losses gives Margenau's Maxwellian [13]. Its tail reaches the 9.9 eV krypton levels. That the energy dependence of ν_m decides how DC and microwave distributions differ is known for discharges in general [17], high-frequency argon distributions have been computed from $\omega \ll \nu$ to $\omega \gg \nu$ [18], and e-e collisions were added for argon discharges [19, 20]. What is new here is the KrF laser gas and its formation chain. A scan repeats the comparison at the absorbed powers per electron of DC fields of 2, 5 and 10 Td in the Ne fill and 3, 6 and 15 Td in the Ar fill, with e-e collisions and ions at each electron density, and with the temporal-growth model of two-term solvers: the electrons grow or decay with the shape of F. Item (3) repeats it with electrons born from the beam. The scan treats the field as given inside the gas. At beam-on density the plasma frequency is 0.47 THz, so a 0.3 THz wave would not enter the gas there. How a field reaches the gas is a separate question of coupling.

(2) Table 3 and Fig. 3 give the result. Electron–electron collisions raise krypton's share in the DC field, from 27.5 to 80.1 % at the power of 2 Td in the Ne fill, as n_e rises from 0 to the beam-on density

2.72×10^{21} m^{-3}. The 28.3 THz field goes from 54.2 to 84.5 %. Collisions bring both towards the Maxwellian, which is where the high frequency was already taking the distribution. At the 9.9 eV krypton threshold, relative to a Maxwellian of the same mean energy, the DC distribution holds 0.556 of the population and the 28.3 THz one 0.742 without e-e collisions, and 0.900 and 0.969 at beam-on density. Over the 30 series of the scan that ratio moves towards 1 between $n_e = 0$ and beam-on density in 29, and stays at 0.94 in the Ar fill at 6 Td and 3 THz. Where the advantage is large, at 2 Td in the Ne fill and at 3 and 6 Td in the Ar fill, the 28.3 THz ratio rises and the DC one rises more. The advantage of the frequency therefore falls from ×1.97 to ×1.055. Normalized by its value without e-e collisions, the advantage falls along nearly one curve of α_g for the three power levels in Fig. 3. It halves at $\alpha_g = 0.41$, 0.46 and 0.55, found by root finding in n_e between the scan's grid points. At beam-on density α_g is 0.023 to 0.14 for these three levels and the frequency is worth 2.0 to 5.5 %. In the Ne fill at the highest power, 10 Td, the high frequency is worse than DC. The fatter tail excites neon itself (its share rises from 1.9 to 7.2 %), and krypton's share falls by 3.0 points. In the Ar fill at 15 Td the high frequency is still 2.9 points better without e-e collisions, and 0.6 points worse at beam-on density.

Table 3: Krypton's share of the absorbed power (%) and the formation efficiency η_{form} of Sec. 8, DC against 28.3 THz at the same absorbed power per electron (that of a DC field of the given E/N without e-e collisions). α_g of the DC point, Eq. (2.5).

fill, power	n_e (m^{-3})	α_g	Kr, DC	Kr, 28.3 THz	ratio	η_{form}, DC	η_{form}, 28.3 THz
Ne, 2 Td	0	–	27.5	54.2	1.973	0.114	0.224
Ne, 2 Td	10^{19}	4.85	29.9	57.6	1.928	0.123	0.237
Ne, 2 Td	10^{20}	0.538	42.7	66.4	1.556	0.176	0.274
Ne, 2 Td	10^{21}	0.0591	70.5	79.4	1.126	0.290	0.325
Ne, 2 Td	2.72×10^{21}	0.0228	80.1	84.5	1.055	0.327	0.344
Ne, 5 Td	0	–	78.5	86.8	1.105	0.322	0.351
Ne, 5 Td	10^{19}	28.7	78.6	86.8	1.104	0.323	0.351
Ne, 5 Td	10^{20}	3.21	79.5	86.9	1.093	0.326	0.351
Ne, 5 Td	10^{21}	0.356	83.2	87.0	1.045	0.339	0.350
Ne, 5 Td	2.72×10^{21}	0.137	85.2	86.9	1.020	0.345	0.349
Ar, 6 Td	0	–	38.4	81.3	2.117	0.149	0.325
Ar, 6 Td	10^{19}	5.74	40.9	83.7	2.047	0.158	0.335
Ar, 6 Td	10^{20}	0.63	53.4	85.3	1.596	0.208	0.342
Ar, 6 Td	10^{21}	0.0676	77.6	87.2	1.124	0.311	0.356
Ar, 6 Td	2.72×10^{21}	0.0259	83.9	87.9	1.048	0.344	0.365

(3) *Beam-sustained electrons.* The growth model lets new electrons appear with the shape of F, and at the lowest power its term carries up to 0.66 of the absorbed power at the points kept, those with $P_{\text{field}} > 1.5\,|P_{\text{growth}}|$. In an e-beam pumped gas the electrons are born from the beam and lost by attachment and by recombination. A second scan therefore repeats the comparison at the absorbed powers of Table 3 with electrons injected with the beam's sub-excitation spectrum (item (6) of Sec. 3) and lost by attachment and by dissociative recombination on $1.075\,n_e$ cations, taken as Kr_2^+ with the rate $2.24 \times 10^{-7}\,T_e^{-1/2}$ cm^3 s^{-1}, built on the dissociative recombination rates of Kannari et al. [31]. The loss has a cross section $\propto 1/\epsilon$, which gives that rate for a Maxwellian, so it takes the slow electrons first. At beam-on density recombination removes electrons 3.1 (Ne

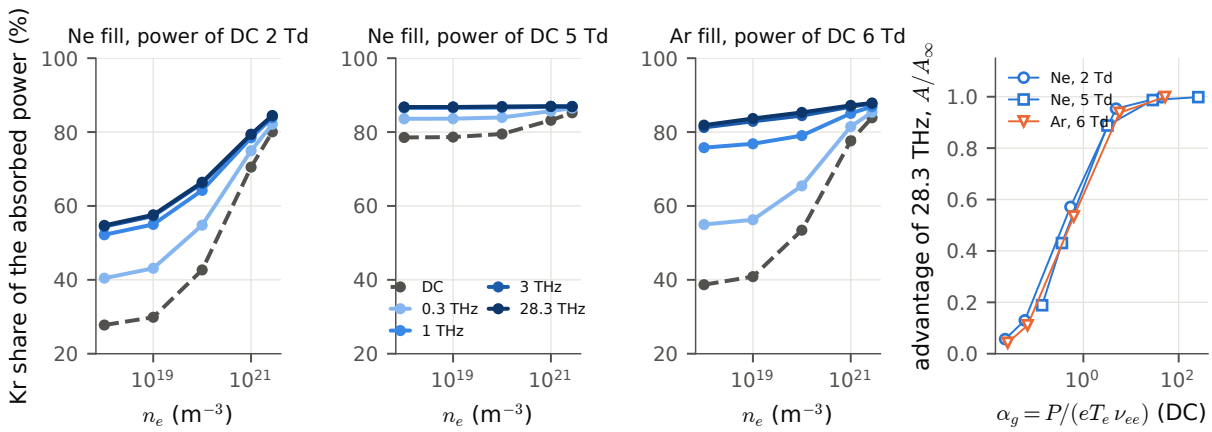


Figure 3: Krypton's share of the absorbed power against n_e, DC (dashed) and 0.3, 1, 3 and 28.3 THz, and (right) the advantage of 28.3 THz over DC, A = ratio -1, normalized by its value without e-e collisions, against α_g. Every point drawn has $P_{\text{field}} > 1.5\,|P_{\text{growth}}|$. The Ar fill at 3 Td is not shown: from 10^{19} m^{-3} up its points at 1 to 28.3 THz fall below that.

fill) and 3.9 (Ar fill) times faster than attachment, and with the DC field the beam's own injected power is then 2.4 to 8.3 times the field's, except at 5 Td in the Ne fill (0.425). Table 4 gives the field's marginal share to krypton at fixed n_e, the change of krypton's power per unit field power. It charges the field with the change of the beam's injected power, which follows the attachment and recombination of the heated distribution. Crediting the beam with its own partition at that power gives the second ratio of the table. Neither ratio holds the beam current fixed. Holding it means finding the electron density at which the beam's injected power per unit volume, and so its current, keeps its value without the field, and the field's absorbed power per unit volume keeps its value at fixed n_e. At beam-on density the advantage at a fixed beam current is within 0.01 of the second ratio. At 10^{18} m^{-3} the field changes the density by a factor $x = 0.36$ to 1.44, and the advantage at a fixed current, $\times 1.23$ to $\times 13.5$, exceeds both ratios. That excess is the field's power per electron, which a fixed absorbed power per unit volume changes by $1/x$: with the field's amplitude held instead, the advantage at a fixed current is the fixed-n_e one within 1 %. The ranges below leave out the points where, with the DC field, the beam injects more power than the field absorbs. At 10^{18} m^{-3} and fixed n_e the frequency raises krypton's share $\times 2.06$ in the Ne fill at 2 Td and $\times 2.40$ in the Ar fill at 6 Td, against $\times 1.97$ and $\times 2.12$ with the growth model, and $\times 11.7$ in the Ar fill at 3 Td, against $\times 27$ with the growth model at the same density. At beam-on density, where only the Ne fill at 5 Td is kept, the advantage is $\times 1.05$ to $\times 1.07$, against $\times 1.020$ with the growth model at the same power, and the field sends 75 to 85 % of its power to krypton, against 85 to 87 %. Where the beam outweighs the field the advantage is $\times 1.10$ to $\times 1.14$ at a fixed beam current and up to $\times 1.27$ at fixed n_e. Counted per change of the total input power, the beam's and the field's together, it is $\times 1.03$ to $\times 1.07$ at every beam-on point. Where the frequency's advantage is large it collapses in both models, and its size at beam-on density depends on how the electrons are created and lost and on how the field's share is counted. Without a field, the e-e collisions at beam-on density send 38.9 % of the injected sub-excitation energy into krypton in the Ne fill and 44.7 % in the Ar fill, against 0.0 % at 10^{18} m^{-3}, and recombination takes 34 to 35 % of it. That is an e-beam effect independent of any field. A loss that takes electrons of every energy alike at the same Maxwellian rate gives 28.5 and 34.2 %, with recombination taking 47 to 49 %, and nearly the

same advantage at 5 Td in the Ne fill, ×1.06 to ×1.07. The beam's own numbers depend on the form of the recombination, and the frequency's advantage much less: at beam-on density each of its four counts moves by at most 0.013 at 5 Td in the Ne fill and 0.044 where the beam outweighs the field.

The beam-on density used throughout, 2.72×10^{21} m^{-3}, is the peak that a full kinetic model of the Ne-fill laser gives under 0.6 MW cm^{-3} of beam. This note's model holds it only with 1.6×10^{30} ion pairs per m^3 and second, 7.9 MW cm^{-3} at the chain's 30.4 eV per pair, which is 13 times that beam, or 2.5 MW cm^{-3} without recombination. Taking every cation as Kr_2^+ overstates the recombination of both fills: Kr^+ forms Kr_2^+ by three-body association in 33 ns in the Ne fill and 17 ns in the Ar fill at that model's 2.5×10^{-31} cm^6 s^{-1}, while Kr_2^+ recombines in 2 ns at beam-on density. That model by default extrapolates its attachment below the floor of its measured tables, and with the measured tables its peak falls to 0.56 of that density. With its beam this note's model holds 5.2×10^{20} m^{-3} in the Ne fill, and there the advantage is ×1.157 to ×1.355 at 2 Td and ×1.051 to ×1.107 at 5 Td in the four counts. That density is a lower bound. Besides overstating recombination, the model injects one electron per ion pair, where the chain of Sec. 8 frees 1.56 through the Penning ionization of krypton by Ne*, and with those it holds 6.2×10^{20} to 7.1×10^{20} m^{-3} at 36.6 to 30.4 eV per pair. Without recombination the advantage at beam-on density is ×1.01 to ×1.18. The density of the beam-pumped gas, and with it what is left of the frequency's advantage, is uncertain by a factor of about five.

Table 4: The beam-sustained model: electrons injected with the beam's sub-excitation spectrum and lost by attachment and recombination, at the absorbed powers of Table 3. Krypton's marginal share $(P_{Kr} - P_{Kr,beam})/P_{field}$ (%) at fixed n_e, the ratio of 28.3 THz over DC with it, with the beam credited with its own partition at its injected power (fixed beam), and at a fixed beam current (see the text), and the formation efficiency of the field's marginal power at fixed n_e. P_{beam}/P_{field} is the beam's injected power over the field's, with the DC field; in the rows marked † the beam outweighs the field.

fill, power	n_e (m^{-3})	P_{beam}/P_{field}	Kr, DC	Kr, 28.3 THz	ratio, fixed n_e	ratio, fixed beam	ratio, fixed current	η_{form}, DC	η_{form}, 28.3 THz
Ne, 2 Td	10^{18}	0.34	30.7	63.4	2.063	2.063	2.647	0.127	0.261
Ne, 2 Td	10^{20}	0.51	42.4	72.4	1.709	1.630	1.633	0.175	0.298
Ne, 2 Td†	2.72×10^{21}	2.85	65.3	75.6	1.157	1.102	1.098	0.265	0.305
Ne, 5 Td	10^{18}	0.05	78.2	88.0	1.125	1.125	1.234	0.321	0.356
Ne, 5 Td	10^{20}	0.06	78.1	87.2	1.117	1.111	1.215	0.320	0.352
Ne, 5 Td	2.72×10^{21}	0.42	75.4	80.8	1.071	1.048	1.052	0.304	0.323
Ar, 3 Td	10^{18}	0.75	5.5	64.5	11.737	11.737	13.515	0.021	0.251
Ar, 3 Td†	10^{20}	1.30	26.0	87.3	3.351	2.597	2.201	0.101	0.343
Ar, 3 Td†	2.72×10^{21}	8.29	60.0	76.2	1.271	1.138	1.135	0.250	0.324
Ar, 6 Td	10^{18}	0.19	39.8	95.5	2.401	2.401	2.777	0.154	0.381
Ar, 6 Td	10^{20}	0.30	49.2	93.4	1.899	1.745	1.769	0.192	0.375
Ar, 6 Td†	2.72×10^{21}	2.45	62.6	76.9	1.227	1.116	1.115	0.262	0.329

7 The self-sustained discharge

(1) A discharge-pumped laser needs ionization to balance attachment. At $\nu_{net} = 0$ the growth model drops out, which makes this the cleanest comparison of field shapes. A third scan finds the

rms field of self-sustainment from DC to $\omega/N = 10^{-11}$ m^3 s^{-1} (78 THz in the Ne fill) for $n_e/N = 0$ to 10^{-4} (Table 5, Fig. 4).

Table 5: The self-sustained discharge: DC, the frequency of the largest krypton share, and $\omega/N = 10^{-11}$ m^3 s^{-1}. P in eV s^{-1} per electron; Kr and buffer shares in %.

fill	n_e/N	ω/N (m^3 s^{-1})	E/N (Td)	$\langle\epsilon\rangle$ (eV)	P	Kr	buffer	η_{form}
Ne	0	DC	8.33	5.31	1.24×10^{10}	87.2	1.0	0.353
Ne	0	3×10^{-14}	10.6	4.94	1.34×10^{10}	88.1	1.8	0.354
Ne	0	10^{-11}	1855	4.06	1.27×10^{10}	86.4	5.2	0.344
Ne	10^{-4}	DC	7.13	4.43	1.13×10^{10}	86.4	3.3	0.345
Ne	10^{-4}	10^{-14}	7.58	4.40	1.14×10^{10}	86.5	3.4	0.345
Ne	10^{-4}	10^{-11}	1769	3.92	1.13×10^{10}	84.9	6.2	0.338
Ar	0	DC	35.9	5.32	2.88×10^{10}	80.5	15.3	0.351
Ar	0	10^{-11}	2507	2.84	3.02×10^{10}	70.2	27.8	0.337
Ar	10^{-4}	DC	30.4	4.86	2.56×10^{10}	78.9	17.0	0.348
Ar	10^{-4}	10^{-11}	2009	3.34	2.39×10^{10}	72.8	24.5	0.340

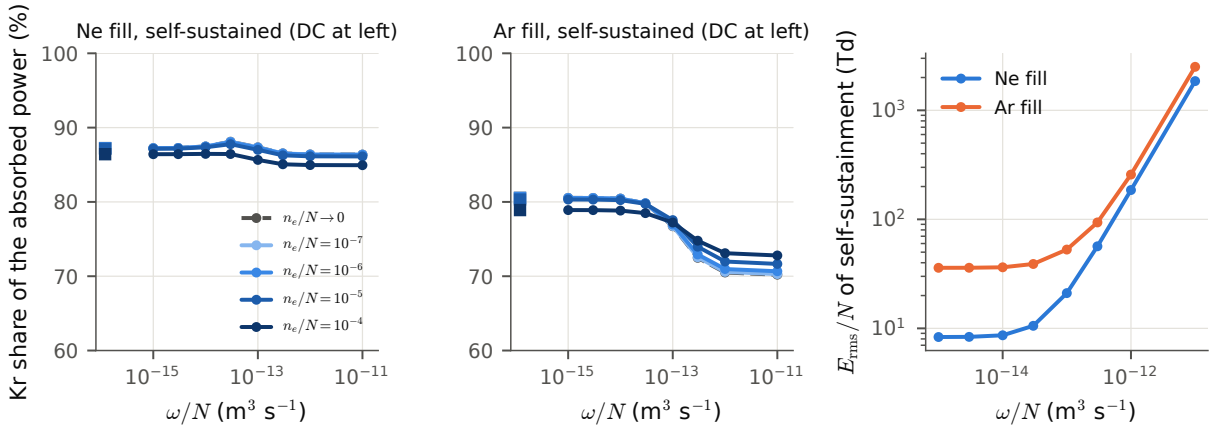


Figure 4: Self-sustained discharges against ω/N: krypton's share in the two fills (DC at the left, squares) and the rms E/N of self-sustainment.

(2) In the Ne fill the discharge sustains itself at 8.33 Td DC with 87.2 % of its power in krypton. On the grid of the scan the share peaks at 88.1 % at $\omega/N = 3 \times 10^{-14}$ m^3 s^{-1}, a third of the Maxwellian mean $\langle\nu_m\rangle/N = 9.2 \times 10^{-14}$ m^3 s^{-1} at the discharge's mean energy: the optimum sits where ω and ν_m are of the same order. A Canon patent on microwave-excited excimer lasers states that at 35 GHz "the plasma excitation efficiency based on identical electric power is highest" at the pressure, 160 Torr, at which the collision frequency of electrons with neon, the main body of the Kr/Ne/F_2 gas, equals the microwave frequency [15]. The optimum exists but is worth 0.9 points. Far above the collision frequency the share falls to 86.4 %, the fatter tail exciting neon. In the Ar fill the discharge needs 35.9 Td DC and gives 80.5 % to krypton, falling to 70.2 % at high frequency as argon's 11.5 eV levels take the tail. The formation efficiency moves less, from 0.351 to 0.337, because argon's excitation partly reaches KrF*. Raising n_e/N to 10^{-4} changes these shares by at

most 2.6 points.

8 What field pumping buys

(1) A formation efficiency turns a power partition into KrF* per absorbed energy,

$$\eta_{\text{form}} = \sum_c f_c \, \frac{h\nu}{u_c} \, b_c, \tag{8.1}$$

with f_c the shares, u_c the energy per event (Kr excitation 10.5 eV, ionization 14.0, Ne 17.5, Ar 12.5), $h\nu = 4.997$ eV, and b_c the branchings of a KrF kinetic network for each fill (Kr* harpoon 0.87 and 0.81, ion channel 0.68). The same accounting gives e-beam pumping of the Ne fill 0.175, from 30.4 eV per ion pair and 0.7 excitations per ion with this fill's $Ne^* \rightarrow Kr^+$ branching 0.80. That accounting gives the beam's sub-excitation electrons nothing. At beam-on density their e-e collisions put 0.159 of their energy into KrF* (Sec. 6, where recombination takes about a third of it), and one such electron of 4.506 eV per 30.4 eV ion pair adds 0.024: the like-for-like e-beam figure there is 0.198. A full kinetic model of the Ne-fill laser, whose Spencer–Fano partition has 36.6 eV per ion pair where the chain counts 30.4, also forms 0.194 of its deposited energy into KrF* at the photon energy, its 1.42 KrF* per ion pair against 1.21 here making up for the larger energy per pair. The chain has no e-beam accounting for the Ar fill.
(2) Field pumping of the beam-ionized gas at beam-on density has $\eta_{\text{form}} = 0.327$ to 0.365 where the field dominates in the growth model (Table 3), 0.327 to 0.349 in the Ne fill, and 0.30 to 0.34 for the field's marginal power with beam-sustained electrons in the Ne fill at 5 Td, in both attributions and at a fixed beam current (Table 4 gives the fixed-n_e values). The self-sustained discharge gives 0.337 to 0.354. In the Ne fill at beam-on density, where the field's power exceeds the beam's own, field pumping thus forms 1.5 to 1.8 times as much as the e-beam's 0.198, and the frequency moves it by −4 to +6 %. At 2 Td, where the beam outweighs the field, the field's marginal power gives 0.30 to 0.33 at a fixed beam current, 1.5 to 1.7 times the e-beam's, and the frequency raises it by 9 % (from 0.265 and by 15 % counted at fixed n_e). This is the rationale of the e-beam-controlled discharge, demonstrated for KrF by Mangano and Jacob [33], and a KrF laser pumped by the discharge that an IR laser beam drives in the gas has also been modelled [21]. Field pumping beats beam pumping at the wall plug when $\eta_{\text{src}}\,\eta_{\text{form,f}} > \eta_{\text{beam}}\,\eta_{\text{form,b}}$, i.e. when the source efficiency exceeds 0.57 to 0.65 of the efficiency with which the pulsed power puts energy into the gas through the foil. The chain's branchings do not depend on n_e. At beam-on density the stepwise ionization $e + Kr^* \rightarrow Kr^+ + 2e$ runs at 0.39 to 1.63 of the harpoon rate in the Ne fill, rising with the power (Kannari's rate at the scan's T_e [31]). It moves Kr* from the harpoon channel (0.87) to the ion channel (0.68), which lowers the KrF* yield of a Kr* by 4.8 to 11.8 %, and so lowers field pumping, which goes mostly through Kr*, more than beam pumping, which goes through ions. In a full kinetic model of microwave co-pumping, the sign of the laser energy that a co-pumping field gains rests on that one rate, and the electron quench of KrF* is a candidate second such rate. The first depends on T_e, and the second may. Those rates, not the frequency, decide whether field pumping pays.

9 At the target

(1) In a direct-drive corona Langdon's $\alpha_L \propto Z_{\text{eff}} I\lambda^2/T_e$ favours short wavelengths. Obenschain et al. computed the quarter-critical temperature and intensity of a 2.6 mm CH sphere driven at 351, 248 and 193 nm (their Table 1, averaged over 2 to 3 ns) [14]. With fully ionized CH, $Z_{\text{eff}} = 37/7$,

Table 6 gives α_L and four estimates of f_L: Langdon's fit, Matte's super-Gaussian, Eq. (3.9) with a constant logarithm, and Eq. (3.9) with the quantum logarithm $L = \ln(2mv^2/\hbar\omega)$ of one electron in both the operator and the absorption. The last is the kinetic counterpart of the improved factor of Turnbull et al. [4], which weights a super-Gaussian with a velocity-dependent Gaunt factor and which Sherlock et al. treat more thoroughly [34]. The e-e logarithm is the NRL formulary's electron–electron value at $n_c/4$ [28], $\ln\Lambda_{ee} = 7.7$, 7.2 and 6.8 at 351, 248 and 193 nm. All of it is for a homogeneous plasma at the time-averaged conditions, with the self-similar distribution at fixed α_L. Nonlocal transport near $n_c/4$ and the structure of the beam are left out.

Table 6: The Langdon factor at the quarter-critical conditions of [14], Table 1. I in 10^{14} W cm^{-2}.

λ (nm)	T_e (keV)	I	α_L	fit	Matte	kinetic, const. L	kinetic, $L(v)$
351	3.88	5.19	0.0325	0.906	0.905	0.908	0.940
248	3.32	5.11	0.0187	0.934	0.933	0.935	0.962
193	2.87	4.44	0.0114	0.953	0.951	0.953	0.975

(2) The kinetic solution with a constant logarithm reproduces the fit within 0.22 %. The velocity-dependent logarithm cuts the penalty $1 - f_L$ to 0.64, 0.58 and 0.52 of the fit's, in line with the finding of Turnbull et al. that the velocity dependence of the Gaunt factor "somewhat mitigates the Langdon effect" [4]. The nonlinear factor alone makes the local absorption rate at quarter critical higher than at 351 nm by ×1.031 (KrF) and ×1.052 (ArF) with Langdon's fit, and by ×1.023 and ×1.038 with the velocity-dependent logarithm. The linear coefficient itself, proportional to $n_c T_e^{-3/2} \propto \lambda^{-2} T_e^{-3/2}$ at quarter critical apart from the logarithm, is ×2.53 and ×5.20 that of 351 nm at the same conditions. The nonlinearity thus adds a few percent to a much larger linear advantage of the excimer drivers. The Langdon effect has been reviewed for laser plasmas [22], computed for short KrF pulses on selenium [23], and quantified against $I\lambda^2$ for the plasmas of extreme-ultraviolet sources, where it lowers the absorption by 10 to 20 % [35]. No published quantification of this wavelength dependence for excimer drivers was found, though not every candidate could be read in full. The bandwidths of Obenschain et al.'s Table 1, 1, 3 and 10 THz, give speckle lifetimes $1/\Delta\nu$ of 1.0, 0.33 and 0.10 ps, shorter than the e-e time $1/\nu_{ee}$ at quarter critical (4.8, 2.0 and 1.0 ps). Whether that changes the Langdon factor of a speckled beam is open. Shaffer et al. found that the Langdon operator predicts the heating well in the simple cases they considered. For a two-colour field it gave slightly less heating than their full calculation, as it did without bandwidth, and a weaker, phase-shifted beat modulation, which they attribute to its steady-state assumption breaking down for broadband light [24]. The scattering of electrons on speckles changes both absorption and transport [25].

10 Verdict and next steps

(1) The nonlinear behavior of IB does not by itself improve a gas laser. The lasing field of an excimer laser stays more than 10^3 times below it. An auxiliary field reaches only the gas nonlinearity, whose effect on the distribution the electron–electron collisions of a pumped laser largely supply: what is left of the frequency's advantage at the beam-on density is 2 to 7 % where the field's power exceeds the beam's own and 10 to 14 % at a fixed beam current where the beam's is larger (1 to 18 % without recombination), and up to ×1.24 at a fixed beam current (×1.355 at fixed n_e) at the five times lower density that this note's electron source and losses give with the beam of the full kinetic

model. At the target the nonlinearity is worth a few percent of absorption rate to KrF and ArF. The ranked next steps, cheapest decisive first:

1. *The instrument.* Fit THz transmission of the laser gas with this solver, which carries the field, the ions and e-e, to read n_e and T_e, and use a THz T_e jump to measure $e + \mathrm{Kr}^* \to \mathrm{Kr}^+ + 2e$ and the electron quench of KrF^*. The first decides the sign of the co-pump in a full kinetic model, the second is the candidate for a second such rate.

2. *Field co-pumping at beam-class power*, frequency-blind: run a full kinetic model of the laser in pairs, with and without the field, with this solver's partition (Tables 3 and 4) and the uncertain rates bracketed, with the electron collisions on Kr^* and KrF^* that the formation chain leaves out.

3. *The e-beam's own sub-excitation electrons.* Item (3) of Sec. 6 sends 39 % of their energy to krypton at beam-on density through e-e collisions in the Ne fill (45 % in the Ar fill), which raises the e-beam's formation efficiency from 0.175 to 0.198. Check whether the energy accounting of full kinetic models of the laser carries it.

4. *The electron source.* The residual advantage at beam-on density depends on how the electrons are created and lost and on how the field's share is counted (1 to 28 %), and on the density itself, which this note's electron source and losses and a full kinetic model put about five times apart. Held at a fixed beam current in the steady state it is within 0.01 of the fixed-beam ratio (Sec. 6). A run in time with the beam's source, attachment, recombination on the fill's own mix of cations and the field together would settle it.

5. *CO_2 lasers.* Of the lasers examined, the one whose own field heats its electrons at the 10 % level, by an order-of-magnitude estimate. Compute it with the discharge's own distribution and timing, and its effect on the vibrational partition and on discharge stability for high-intensity amplifiers.

6. *The target.* Use the velocity-dependent Langdon factor in the wavelength studies of excimer drivers, and test whether THz bandwidth and speckle change it.

A The electron–electron operator in energy

(1) The isotropic Rosenbluth operator for electrons of distribution $f(v)$, $\int 4\pi v^2 f \,\mathrm{d}v = n_e$, is [10]

$$\frac{\partial f}{\partial t} = \frac{4\pi\Gamma}{v^2}\frac{\partial}{\partial v}\Big[f\, I_2 + \frac{I_4 + v^3 J_1}{3v}\frac{\partial f}{\partial v}\Big], \qquad \Gamma = \frac{e^4 \ln\Lambda}{4\pi\epsilon_0^2 m^2}, \tag{A.1}$$

with $I_2 = \int_0^v u^2 f$, $I_4 = \int_0^v u^4 f$, $J_1 = \int_v^\infty u f$. For a Maxwellian $J_1 = v_t^2 f$ and $I_4 = 3v_t^2 I_2 - v^3 v_t^2 f$, so the bracket vanishes.
(2) Write $h(v,w) = \min(v,w)^3/3$. Since $\partial_w h = w^2$ for $w < v$ and 0 above, $-\int f'(w) h \,\mathrm{d}w = I_2$. And $(I_4 + v^3 J_1)/3v = \int f(w)(w/v) h \,\mathrm{d}w$. The bracket is therefore

$$\int_0^\infty w\, h(v,w)\Big[f(w)\frac{f'(v)}{v} - f(v)\frac{f'(w)}{w}\Big]\mathrm{d}w, \tag{A.2}$$

so v times the bracket has the symmetric weight $v\,w\,h$ and an antisymmetric bracket, which is the Landau form. In energy, $f'(v)/v = (m/e)\,\partial_\epsilon f$ and $w\,\mathrm{d}w = (e/m)\mathrm{d}\epsilon'$, and with $f = n_e(2\pi)^{-1}(m/2e)^{3/2}F$ the upward number flux per electron becomes Eq. (3.6).

(3) Integrating by parts in ϵ', $\int \min^{3/2}[F(\epsilon')F' - FF'(\epsilon')]\mathrm{d}\epsilon' = F'(A_2 + \epsilon^{3/2}A_3) + \frac{3}{2}A_1F$, which is half the bracket of Hagelaar and Pitchford, so their a [9] equals $(2\Gamma/3)(m/2e)^{3/2} = e^2\gamma \ln\Lambda/24\pi\epsilon_0^2$, as written in Eq. (3.6).

B The checks

A check script computes every result, or reloads it from the stored output of the scan that made it and recomputes a sample of that output, and tests it. It recomputes the computed numbers of the prose to their printed digits, each inside a sentence that occurs once in the text, compares the inputs the text states with the code that uses them, and tests the worded claims against the data. It re-derives the displayed formulas numerically and requires them in the text as printed. The numbers quoted from sources were checked against the sources by hand. Table 7 counts the checks by the section they support.

Table 7: The checks of these notes, by the section whose statements, equations, tables and figures they support: all checks, the computed numbers of the prose recomputed to their printed digits, and the worded claims tested against the data.

supports	checks	printed numbers	claims
Sec. 1	58	40	16
Sec. 2	18	8	1
Sec. 3	46	12	5
Sec. 4	20	8	3
Sec. 5	45	20	19
Sec. 6	150	95	30
Sec. 7	23	14	5
Sec. 8	61	47	4
Sec. 9	29	18	4
Sec. 10	17	14	3
Appendix A	1		
Appendix B	2		
the references and the text as a whole	65		
Total	535	276	90

References

[1] A. B. Langdon, Nonlinear inverse bremsstrahlung and heated-electron distributions, Phys. Rev. Lett. **44**, 575 (1980), doi:10.1103/PhysRevLett.44.575.

[2] J. P. Matte, M. Lamoureux, C. Möller, R. Y. Yin, J. Delettrez, J. Virmont and T. W. Johnston, Non-Maxwellian electron distributions and continuum X-ray emission in inverse bremsstrahlung heated plasmas, Plasma Phys. Control. Fusion **30**, 1665 (1988), doi:10.1088/0741-3335/30/12/004.

[3] D. Turnbull et al., Inverse bremsstrahlung absorption, Phys. Rev. Lett. **130**, 145103 (2023), doi:10.1103/PhysRevLett.130.145103.

[4] D. Turnbull et al., Reconciling calculations and measurements of inverse bremsstrahlung absorption, Phys. Plasmas **31**, 063304 (2024), doi:10.1063/5.0203446.

[5] A. L. Milder et al., Measurements of non-Maxwellian electron distribution functions and their effect on laser heating, Phys. Rev. Lett. **127**, 015001 (2021), doi:10.1103/PhysRevLett.127.015001.

[6] C. K. Watson et al., Unified microscopic foundations of the quantum Boltzmann equation, Fluct. Noise Lett. **25**, 2640002 (2026), doi:10.1142/S021947752640002X.

[7] S. Atzeni and J. Meyer-ter-Vehn, *The Physics of Inertial Fusion* (Oxford University Press, Oxford, 2004), doi:10.1093/acprof:oso/9780198562641.001.0001.

[8] M. Yu. Glyavin, A. G. Luchinin, G. S. Nusinovich, J. Rodgers, D. G. Kashyn, C. A. Romero-Talamas and R. Pu, A 670 GHz gyrotron with record power and efficiency, Appl. Phys. Lett. **101**, 153503 (2012), doi:10.1063/1.4757290.

[9] G. J. M. Hagelaar and L. C. Pitchford, Solving the Boltzmann equation to obtain electron transport coefficients and rate coefficients for fluid models, Plasma Sources Sci. Technol. **14**, 722 (2005), doi:10.1088/0963-0252/14/4/011.

[10] M. N. Rosenbluth, W. M. MacDonald and D. L. Judd, Fokker–Planck equation for an inverse-square force, Phys. Rev. **107**, 1 (1957), doi:10.1103/PhysRev.107.1.

[11] H. F. Walker and P. Ni, Anderson acceleration for fixed-point iterations, SIAM J. Numer. Anal. **49**, 1715 (2011), doi:10.1137/10078356X.

[12] M. J. Druyvesteyn and F. M. Penning, The mechanism of electrical discharges in gases of low pressure, Rev. Mod. Phys. **12**, 87 (1940), doi:10.1103/RevModPhys.12.87.

[13] H. Margenau, Conduction and dispersion of ionized gases at high frequencies, Phys. Rev. **69**, 508 (1946), doi:10.1103/PhysRev.69.508.

[14] S. P. Obenschain et al., Direct drive with the argon fluoride laser as a path to high fusion gain with sub-megajoule laser energy, Phil. Trans. R. Soc. A **378**, 20200031 (2020), doi:10.1098/rsta.2020.0031.

[15] T. Ohmi, N. Tanaka and M. Hirayama (Canon Inc.), Excimer laser oscillation apparatus and method, excimer laser exposure apparatus, and laser tube, US Patent 6,331,994 B1 (published 2001-12-18).

[16] V. P. Silin, Nonlinear high-frequency plasma conductivity, Sov. Phys. JETP **20**, 1510 (1965) [Zh. Eksp. Teor. Fiz. **47**, 2254 (1964)], record at http://www.jetp.ras.ru/cgi-bin/e/index/e/20/6/p1510?a=list.

[17] E. V. Karoulina and Yu. A. Lebedev, The influence of the electron transport cross sectional shape on electron energy distribution functions in DC and microwave plasmas, J. Phys. D **21**, 411 (1988), doi:10.1088/0022-3727/21/3/006.

[18] C. M. Ferreira and J. Loureiro, Electron energy distributions and excitation rates in high-frequency argon discharges, J. Phys. D **16**, 2471 (1983), doi:10.1088/0022-3727/16/12/024.

[19] P. A. Sá, J. Loureiro and C. M. Ferreira, Effects of electron–electron collisions on the characteristics of DC and microwave discharges in argon at low pressures, J. Phys. D **25**, 960 (1992), doi:10.1088/0022-3727/25/6/010.

[20] D. Wang, T. Ma and B. Guo, Electron energy distributions and kinetic temperatures in high-frequency argon discharges, J. Phys. D **26**, 62 (1993), doi:10.1088/0022-3727/26/1/010.

[21] V. I. Igoshin and S. Yu. Pichugin, Pumping a KrF excimer laser by a laser discharge induced by an IR laser beam, Quantum Electron. **23**, 33 (1993), doi:10.1070/QE1993v023n01ABEH002931.

[22] N. R. Shaffer, T. Seo, S. X. Hu and V. N. Goncharov, The Langdon effect in laser plasmas: Absorption and conduction, Phys. Plasmas **32**, 062703 (2025), doi:10.1063/5.0266182.

[23] P. E. Pulsifer and K. G. Whitney, Fokker–Planck analysis of short-pulse laser heating in the hot-spot-model approximation, Phys. Rev. E **50**, 4926 (1994), doi:10.1103/PhysRevE.50.4926.

[24] N. R. Shaffer, M. Sherlock, A. V. Maximov and V. N. Goncharov, An extended Vlasov–Fokker–Planck approach for kinetic simulations of laser plasmas, Phys. Plasmas **30**, 043906 (2023), doi:10.1063/5.0143248.

[25] M. Sherlock and P. Michel, Absorption and transport effects induced in plasmas by the interaction of electrons with laser speckles, Phys. Rev. Lett. **129**, 215001 (2022), doi:10.1103/PhysRevLett.129.215001.

[26] M. Tachikawa, H. Murakami, F. O. Shimizu and T. Shimizu, Novel aspects of the optogalvanic effect in the CO_2 laser medium, Jpn. J. Appl. Phys. **29**, 958 (1990), doi:10.1143/JJAP.29.958.

[27] E. Carbone et al., Data needs for modeling low-temperature non-equilibrium plasmas: the LXCat project, history, perspectives and a tutorial, Atoms **9**, 16 (2021), doi:10.3390/atoms9010016; the BSR, Biagi-v7.1, IST-Lisbon and SIGLO databases at www.lxcat.net.

[28] A. S. Richardson, *2019 NRL Plasma Formulary* (Naval Research Laboratory, Washington, DC, 2019).

[29] V. L. Ginzburg and A. V. Gurevich, Nonlinear phenomena in a plasma located in an alternating electromagnetic field, Sov. Phys. Usp. **3**, 115 and 175 (1960), doi:10.1070/PU1960v003n01ABEH003261 and doi:10.1070/PU1960v003n02ABEH003267.

[30] A. V. Gurevich, *Nonlinear Phenomena in the Ionosphere* (Springer, Berlin, 1978), doi:10.1007/978-3-642-87649-3.

[31] F. Kannari, M. Obara and T. Fujioka, An advanced kinetic model of electron-beam-excited KrF lasers including the vibrational relaxation in KrF^*(B) and collisional mixing of KrF^*(B,C), J. Appl. Phys. **57**, 4309 (1985), doi:10.1063/1.334590.

[32] K. Sakamoto et al., Achievement of robust high-efficiency 1 MW oscillation in the hard-self-excitation region by a 170 GHz continuous-wave gyrotron, Nature Phys. **3**, 411 (2007), doi:10.1038/nphys599.

[33] J. A. Mangano and J. H. Jacob, Electron-beam-controlled discharge pumping of the KrF laser, Appl. Phys. Lett. **27**, 495 (1975), doi:10.1063/1.88547.

[34] M. Sherlock, P. Michel, D. J. Strozzi, L. Divol, E. Kur and G. Zimmerman, Inverse bremsstrahlung absorption rate for super-Gaussian electron distribution functions including plasma screening, Phys. Rev. E **109**, 055201 (2024), doi:10.1103/PhysRevE.109.055201.

[35] J. Gonzalez and J. Sheil, Langdon effect in the realm of extreme ultraviolet source plasmas, Phys. Rev. E **111**, L023201 (2025), doi:10.1103/PhysRevE.111.L023201.